

1. Consider a function $f(x) = x^3 + x^2 - 4x - 4$.

(a) (1 marks) Compute the minimum number of iterations required to find the root within the interval $[-10, -1.5]$ if the machine epsilon (error bound) is 1×10^{-2} .

(b) (4 marks) Show 5 iterations using the Bisection Method to find the root of the above function within the interval $[-10, -1.5]$.

(c) (2 marks) State the exact roots of $f(x)$ and construct two different fixed point functions $g(x)$ such that $f(x) = 0$.

(d) (3 marks) Compute the convergence rate of each fixed point function $g(x)$ obtained in the previous part, and state which root it is converging to or diverging.

②

$$n \gg \frac{\log(1 - 1.5 + 10) - \log(1 \times 10^{-2})}{\log 2} \approx 9$$

$$\therefore n \gg 879 \approx 9$$

⑥

$$x_l = -10, x_u = -1.5$$

$$\text{Ite-1} \quad x_m = \frac{x_l + x_u}{2} = \frac{-10 + (-1.5)}{2} = -5.75$$

$$f(x_l) \times f(x_m) = f(-10) \times f(-5.75) \approx 119.27 \times 10^3$$

$$\therefore x_l = x_m, x_u = x_u$$

Ite-2

$$x_m = \frac{-5.75 + (-1.5)}{2} = -3.625$$

$$f(x_l) \times f(x_m) = f(-5.75) \times f(-3.625) = 3.31 \times 10^3$$

$$\therefore x_l = x_m, x_u = x_u$$

Ite-3

$$x_m = \frac{-3.625 + (-1.5)}{2} = -2.5625$$

$$f(x_l) \times f(x_m) = f(-3.625) \times f(-2.5625) = 96.217$$

$$\therefore x_l = x_m, x_u = x_u$$

↓ Iteration continues like this.

③ $f(x) = x^3 - x^2 - 4x - 4$

Solving $f(x) = 0$, we get,

$$x_1 = -1$$

$$x_2 = -2$$

$$x_3 = 2$$

$$\begin{aligned} \therefore f(x) &= 0 \\ \Rightarrow x^3 + x^2 - 4x - 4 &= 0 \\ \Rightarrow x^2 &= -x^3 + 4x + 4 \\ \Rightarrow x &= \sqrt{-x^3 + 4x + 4} \\ \Rightarrow g_1(x) &= \sqrt{-x^3 + 4x + 4} \end{aligned}$$

$$\begin{aligned} \therefore f(x) &= 0 \\ \Rightarrow x^3 + x^2 - 4x - 4 &= 0 \\ \Rightarrow 4x &= x^3 + x^2 - 4 \\ \Rightarrow x &= \frac{x^3 + x^2 - 4}{4} \\ \Rightarrow g_2(x) &= \frac{x^3 + x^2 - 4}{4} \end{aligned}$$

④ Here,

$$\lambda_1 = |g_1'(root)|$$

$$\lambda_2 = |g_2'(root)|$$

$$\begin{aligned} g_1(x) &= \sqrt{-x^3 + 4x + 4} & g_2(x) &= \frac{x^3 + x^2 - 4}{4} \\ \therefore g_1'(x) &= \frac{4 - 3x^2}{2\sqrt{-x^3 + 4x + 4}} & g_2'(x) &= \frac{3x^2 + 2x}{4} \end{aligned}$$

Now, for roots -1, -2 and 2, we get,

$$\lambda_1 = |g_1'(-1)| = 0.5 \quad [0 \leq \lambda < 1]$$

$$\lambda_1 = |g_1'(-2)| = 2 \quad [\lambda > 1]$$

$$\lambda_1 = |g_1'(2)| = 2 \quad [\lambda > 1]$$

$$\lambda_2 = |g_2'(-1)| = 0.25 \quad [0 \leq \lambda < 1]$$

$$\lambda_2 = |g_2'(-2)| = 2 \quad [\lambda > 1]$$

$$\lambda_2 = |g_2'(2)| = 4 \quad [\lambda > 1]$$

Root -1 is converging for both $g_1(x)$ and roots -2 and 2 are diverging.

2. Consider the following function: $f(x) = xe^x - 1$.

(a) (4 marks) Find solution of $f(x) = 0$ up to 5 iterations using Newton's method starting with $x_0 = 1.5$. Keep up to four significant figures.

(b) (4 marks) Consider the fixed point function, $g(x) = \frac{2x+1}{\sqrt{x+1}}$. Show that to be super-linearly convergent, the root must satisfy $x^* = \frac{-3}{2}$.

② $f(x) = xe^x - 1$ and $x_0 = 1.5$ $\therefore f'(x) = e^x + xe^x$

Iteration-1

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.5 - \frac{f(1.5)}{f'(1.5)} = 0.9891$$

Iteration -2

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.9891 - \frac{f(0.9891)}{f'(0.9891)} = 0.6787$$

Iteration-3

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.6787 - \frac{f(0.6787)}{f'(0.6787)} = 0.5766$$

↓ Iteration continues like this.

⑥ Given, $g(x) = \frac{2x+1}{\sqrt{x+1}}$

$$\therefore g'(x) = \frac{2x+3}{2(x+1)^{3/2}}$$

To be super linear convergent, $\lambda = 0$.

$$\therefore \lambda = |g'(x)| = 0$$

$$\Rightarrow \frac{2x+3}{2(x+1)^{3/2}} = 0$$

$$\Rightarrow 2x + 3 = 0 \quad \text{or, } x = -3/2 \text{ (showed)}$$

3. (a) (4 marks) Consider a cubic function, $f(x) = 2x^3 - 2x - 5$. Compute a **superlinearly convergent fixed point function $g(x)$** for the given function $f(x)$ using **Newton's method**.

(b) (4 marks) Consider the function $f(x) = \cos(2x) - \sin(x)$. Compute the root of the function using Newton's method with Aitken's acceleration and starting point, $x_0 = 0$. Consider up to five decimal places. [Error bound is 1×10^{-4}] [You must use Radian Mode of calculator]

② Given, $f(x) = 2x^3 - 2x - 5$ $\therefore f'(x) = 6x^2 - 2$

Using newton's method,

$$g(x) = x - \frac{f(x)}{f'(x)}$$
$$= x - \frac{2x^3 - 2x - 5}{6x^2 - 2}$$
$$= \frac{6x^3 - 2x - 2x^3 + 2x + 5}{6x^2 - 2}$$
$$= \frac{4x^3 + 5}{6x^2 - 2}$$

⑥ Given, $f(x) = \cos(2x) - \sin x$. $\therefore f'(x) = -2\sin 2x - \cos x$

$$\epsilon = 1 \times 10^{-3}, x_0 = 0$$

$$x = \frac{f(x_0)}{f'(x_0)} = 0 - \frac{f(0)}{f'(0)} = 1$$

Given, $f(x) = x^3 - x^2 - 1$

$$\epsilon = 1 \times 10^{-3}, x_0 = 0$$

k	x_k	$f(x_k)$	$ f(x_k) < \epsilon$
0	0	1	No
1	1	-1.25761	No
2	0.46686	0.14476	No
$\hat{2}$	0.65226	-0.34384	No
3	0.52604	-0.00695	No
4	0.52366	-0.0000028	Yes

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{f(0)}{f'(0)} = 1$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1 - \frac{f(1)}{f'(1)} = 0.46686$$

$$\hat{x}_2 = x_0 - \frac{(x_1 - x_0)^2}{x_2 - 2x_1 + x_0} = 0 - \frac{(1-0)^2}{0.46686 - 2 + 0} = 0.65226$$

$$x_3 = \hat{x}_2 - \frac{f(\hat{x}_2)}{f'(\hat{x}_2)} = 0.52604$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.52366$$

4. In the interval $[-4, 4]$, the function, $f(x) = x^3 - x^2 - 3x + 2$, has three roots at 2.000, 0.6180 and -1.618; and two turning points at $x = -0.721$ and $x = 1.387$.

a) (1 mark) Explain why it might not be possible to find the root of the given function and interval using the interval bisection method.

b) (3 marks) Write down the correct intervals, including the root it contains, such that the problem in the previous part can be avoided.

⊙ Bisection method works for such interval where only one root exists. In above question, within $[-4, 4]$, there are two turning points which means multiple root exists there. As such, it might not be possible to find root.

ⓑ Correct intervals will be.

$$[-4, -0.721] \cup [-0.721, 1.387] \cup [1.387, 4]$$

Interval $[-4, -0.721]$ will contain root -1.618

" $[-0.721, 1.387]$ " " " 0.6180

" $[1.387, 4]$ " " " 2